



Expectation and reality

EXPECTATION IS AN OUTCOME THAT IS ANTICIPATED TO OCCUR;
REALITY IS THE OUTCOME THAT ACTUALLY OCCURS.

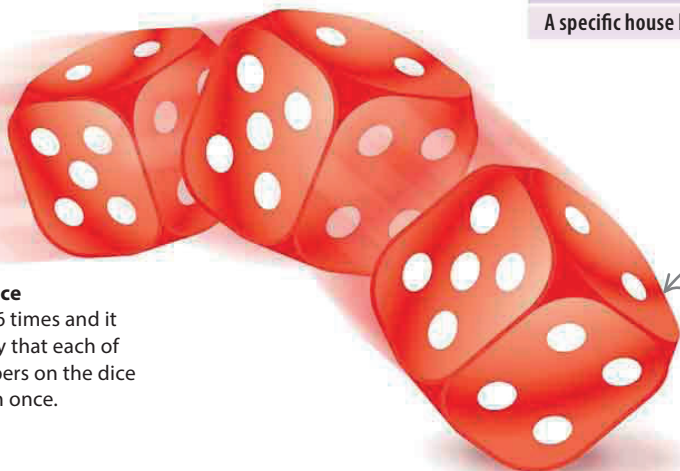
The difference between what is expected to occur and what actually occurs can often be considerable.

What is expectation?

There is an equal chance of a 6-sided dice landing on any number. It is expected that each of the 6 numbers on it will be rolled once in every 6 throws ($\frac{1}{6}$ of the time). Similarly, if a coin is tossed twice, it is expected that it will land on heads once and tails once. However, this does not always happen in real life.

WHAT ARE THE CHANCES?

Two random phone numbers ending in same digit	1 chance in 10
Randomly selected person being left-handed	1 chance in 12
Pregnant woman giving birth to twins	1 chance in 33
An adult living to 100	1 chance in 50
A random clover having four leaves	1 chance in 10,000
Being struck by lightning in a year	1 chance in 2.5 million
A specific house being hit by a meteor	1 chance in 182 trillion



chance of rolling each number is 1 in 6

△ Roll a dice

Roll a dice 6 times and it seems likely that each of the 6 numbers on the dice will be seen once.

Expectation versus reality

Mathematical probability expects that when a dice is rolled 6 times, the numbers 1, 2, 3, 4, 5, and 6 will appear once each, but it is unlikely this outcome would actually occur. However, over a longer series of events, for example, throwing a dice a thousand times, the total numbers of 1s, 2s, 3s, 4s, 5s, and 6s thrown would be more even.

▷ Expectation

Mathematical probability expects that, when a dice is rolled 6 times, a 4 will be thrown once.

▷ Reality

Throwing a dice 6 times may create any combination of the numbers on a dice.

